Jiawen Gong

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Education

University of Illinois Urbana–Champaign Expected May 2028
B.S. in Mathematics & Computer Science GPA: 3.9/4.0

Work Experience

Alibaba Group (Taobao & Tmall) — *AI Agent Engineer Intern* **Hangzhou · Jun–Aug 2026**

Worked on post-training, tooling, and evaluation for TMALL AI Assistant, an e-commerce agent with **1M+ daily uses**; in production, task success improved by **28.4%** and completion time decreased by **49.5%**.

- Designed and developed the post-training pipeline: built reproducible commerce environments, generated execution-verified multi-turn, multi-tool trajectories from 9,600 tasks, and tuned the agent with SFT and DPO; offline task success improved from **43.8% to 78.9%** on a fixed held-out evaluation.
- Developed tools and skills for a system with **120 API/GUI tools and 30+ execution flows**; implemented a multi-agent graph for goal decomposition, domain routing, and tool-driven replanning, using page, cart, and order state to control graph transitions and termination.
- Built a **2,400-case stratified evaluation suite** with deterministic business-state checks and an LLM judge validated against human labels ($\kappa = 0.83$); prioritized data, tool, orchestration, and runtime fixes via root-cause analysis.

Baidu Inc. (Baidu AI Cloud) — *Software Engineer Intern* **Remote · Mar–May 2026**

Built a RAG system over database documentation to power data-agent Q&A and troubleshooting.

- Processed **80,000 documents** into chunked indices and a normalized entity-relation graph; designed vector/BM25 recall, RRF fusion, and graph expansion for multi-hop retrieval.
- Improved **Recall@5 from 71.8% to 89.6%** and MRR from 0.62 to 0.78 at sub-1.8-second p95 latency; exposed the pipeline as a RAG search tool for agents to search, refine queries, and retrieve iteratively.

University of Illinois Urbana–Champaign — *Course Assistant, CS 277* **Jan 2026–Present**

- Led labs, office hours, grading, and code reviews for a 400-student data structures course; built PrairieLearn auto-graders for 10,000+ submissions and reduced platform errors by **40%** through regression testing.

Research & Projects

Your Benchmark Is an Empirical Measure Over Difficulty
Co-first author **NeurIPS 2026**

- Formalized benchmarks as empirical measures over item difficulty and derived conditions for model-ranking stability and localized-progress detection; an audit of **25 benchmarks across seven domains** revealed ranking instability from difficulty skew and coverage gaps that obscured localized training gains.
- Developed BENCHFILL using difficulty-conditioned rewriting, online IRT calibration, and memory-guided evolutionary search; reduced the sum of the three largest coverage gaps by **32.67% vs. 17.52%** for a budget-matched baseline.
- Expanded an AIME24–26 evaluation set from 30 to 60 items; the augmented benchmark detected all three localized post-SFT gains missed by the original and budget-matched baselines (**3/3 vs. 0/3**, $p < 0.05$).

MonitorBench: A Comprehensive Benchmark for Chain-of-Thought Monitorability in Large Language Models
Co-author **COLM 2026**

- Built an open-source benchmark of **1,514 instances** spanning 19 tasks and seven categories; developed causal-effect-adjusted metrics and stress tests for concealment and monitor-aware evasion.
- Evaluated eight LLMs: CoT monitoring outperformed action-only monitoring by **24 points**, while monitorability correlated negatively with model capability ($r = -0.87$ to -0.39) and dropped by 27 points when task-critical factors affected only final answers.

SWE-Bench Software Repair Agent **Sep–Dec 2025**

- Built a repair agent that generated minimal, test-validated patches from bug reports via Dockerized reproduce-edit-test loops; exposed repository search, editing, and testing through typed MCP calls for deterministic, auditable runs.
- Achieved **34.8% Pass@1** on SWE-Bench Verified, up from 4.2% for the baseline agent.

Technical Skills

Languages: Python, C++, Java, Kotlin, JavaScript, SQL
ML & Agents: PyTorch, LangGraph, NumPy, RAG, SFT, GRPO, DPO, Megatron-LM, LLM Evaluation, MCP
Systems & Tools: Linux, Docker, Git, AWS (EC2), MySQL, Node.js